Can combine:

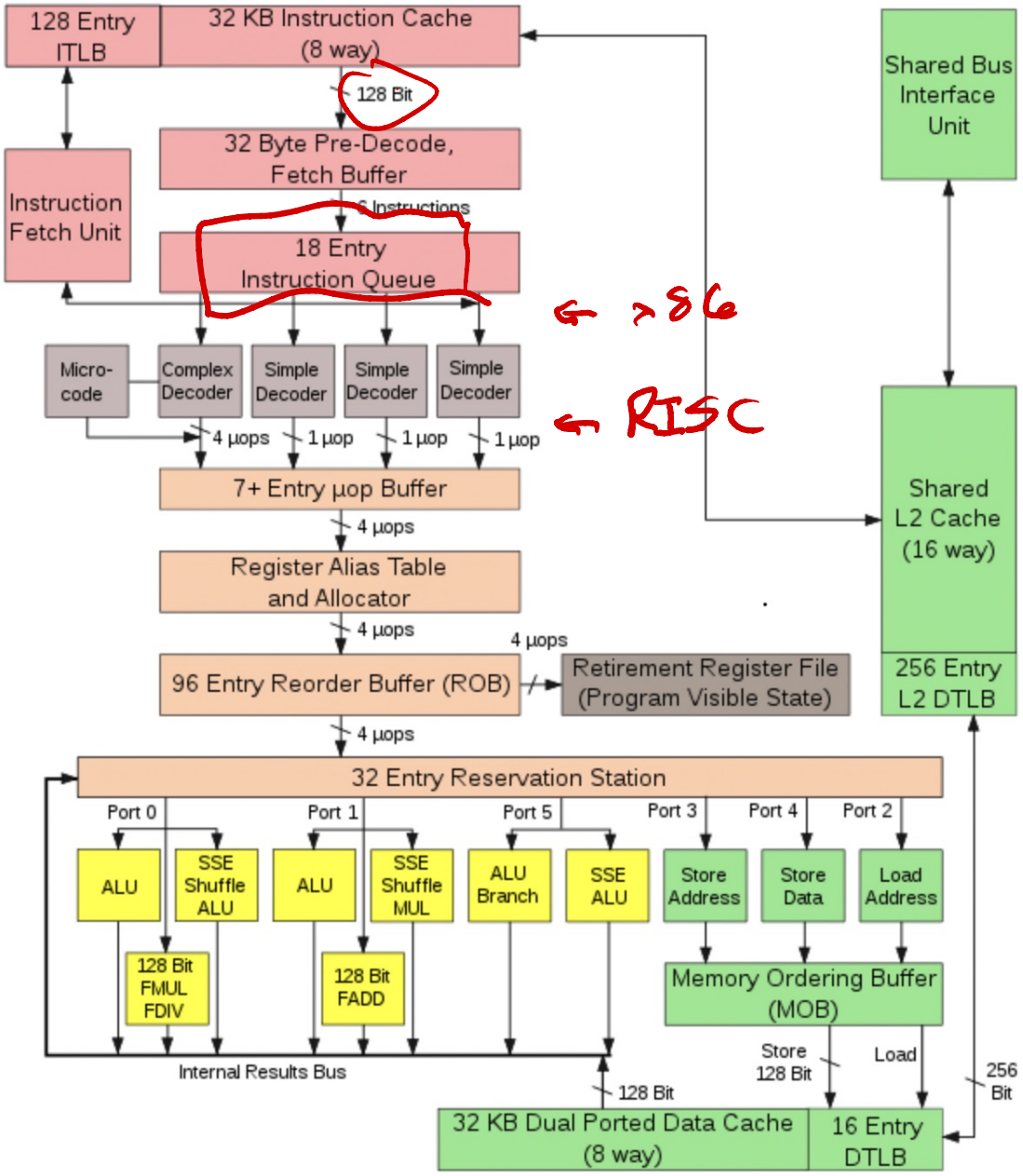
Dispatch queue

w/

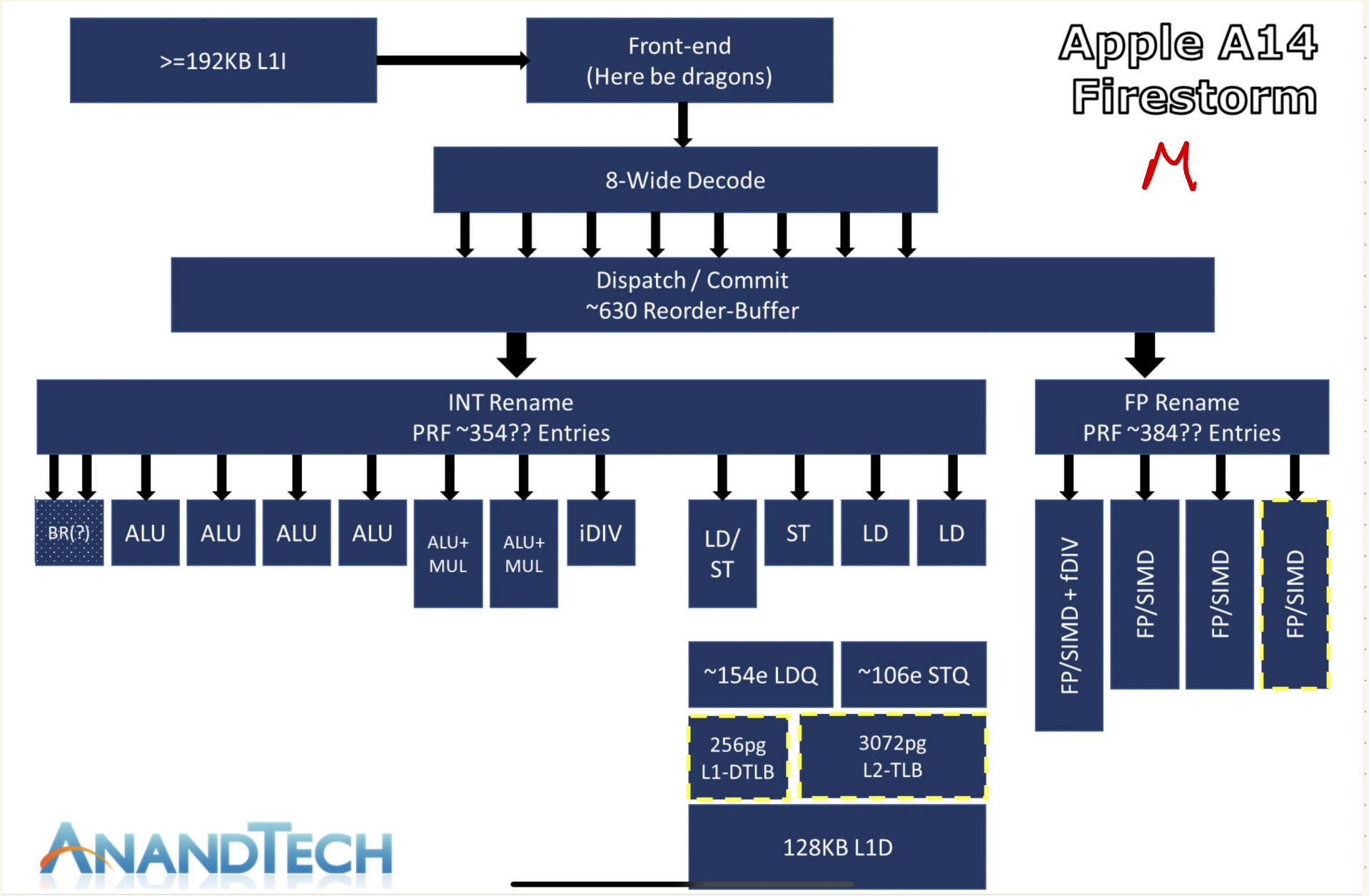
Sched queue

w/

ROB



Intel Core 2 Architecture


Apple A14
Firestorm
>=192KB L1I
Front-end
(Here be dragons)
8-Wide Decode
Dispatch / Commit
~630 Reorder-Buffer
INT Rename
PRF ~354?? Entries
FP Rename
PRF ~384?? Entries
BR(?)
ALU
ALU
ALU
ALU
ALU+
MUL
ALU+
MUL
iDIV
LD/
ST
ST
LD
LD
FP/SIMD + fDIV
FP/SIMD
FP/SIMD
FP/SIMD
~154e LDQ
~106e STQ
256pg
L1-DTLB
3072pg
L2-TLB
128KB L1D
ANANDTECH

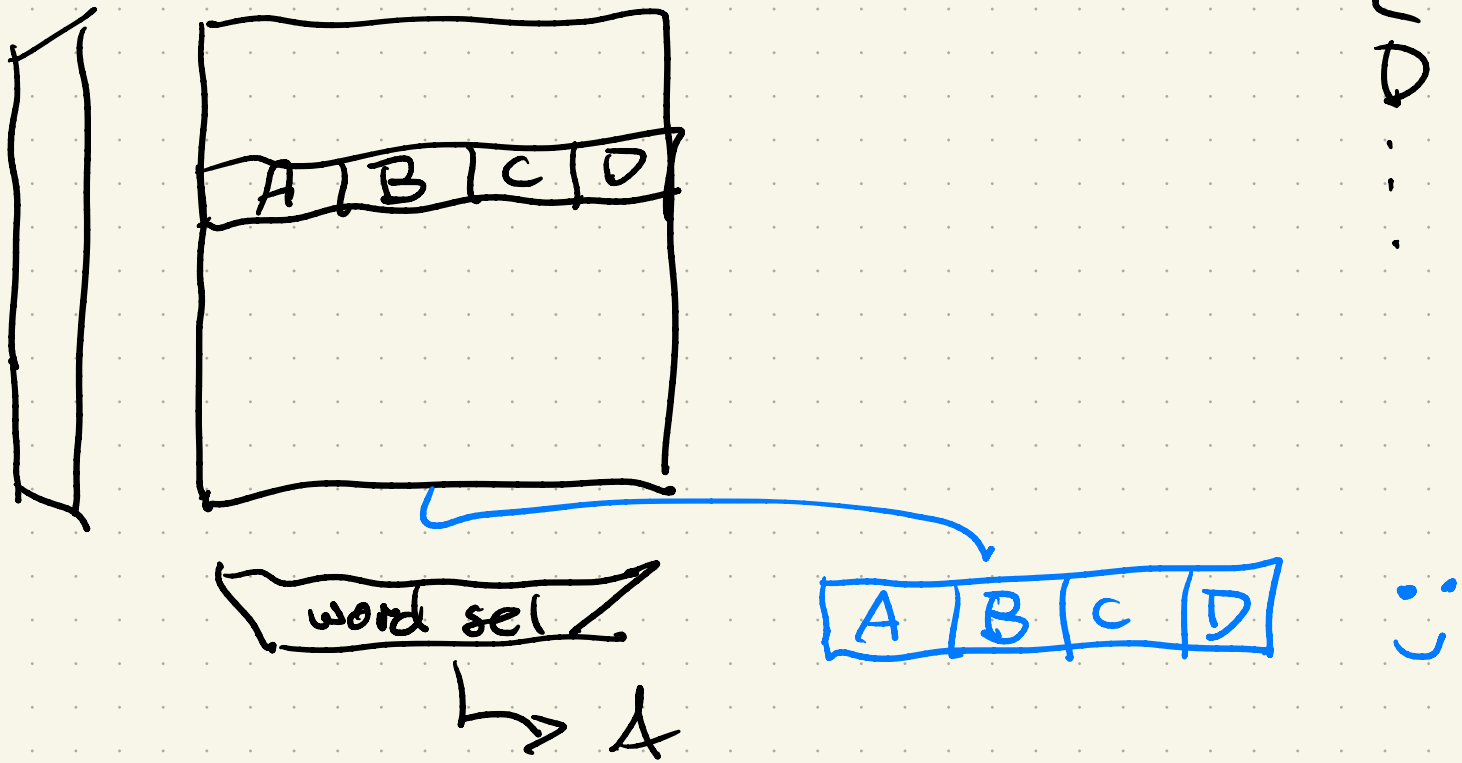
Retired IPC = Min (Fetched IPC,
Dispatched IPC, Issued (Fired) IPC,
completed IPC)

I cache 4 inst/cycle

PC → I$ → 1 instruction

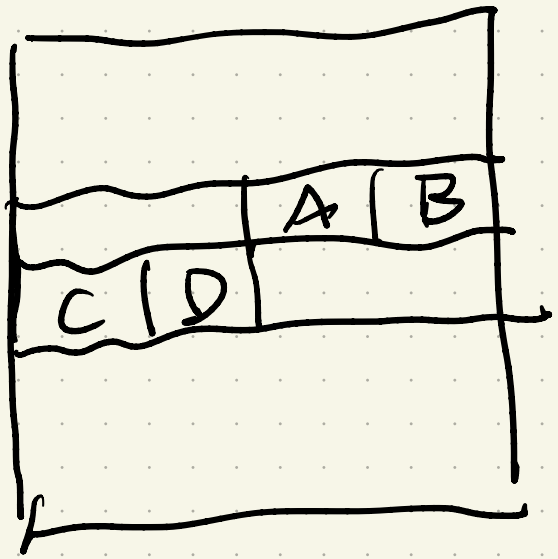
Assume ISA fixed 4-byte inst.

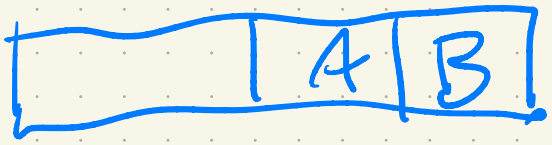
Next PC = A



A
B
C
D
...

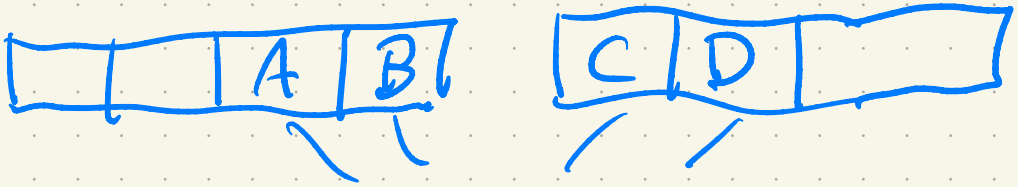
Nextpc = A



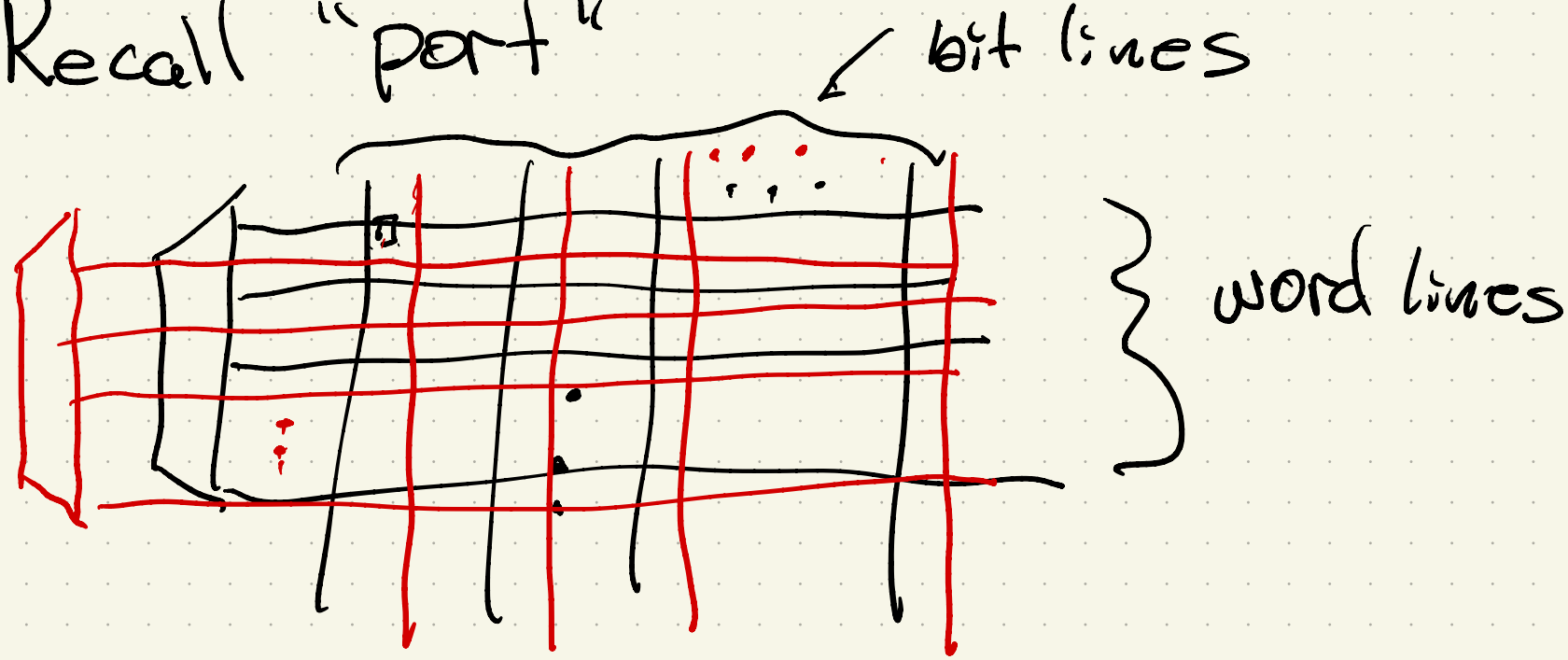


fetch rate = 2 :"

Pull down 2 blocks => 2 ports



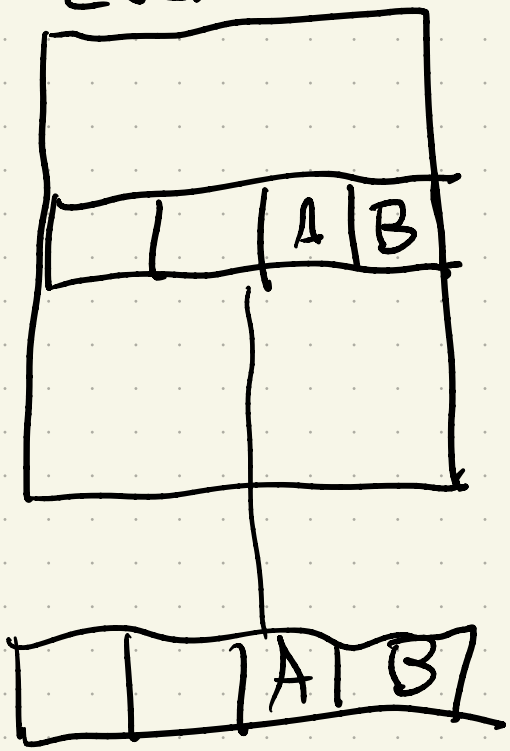
Recall "port"



⇒ Dual porting is expensive

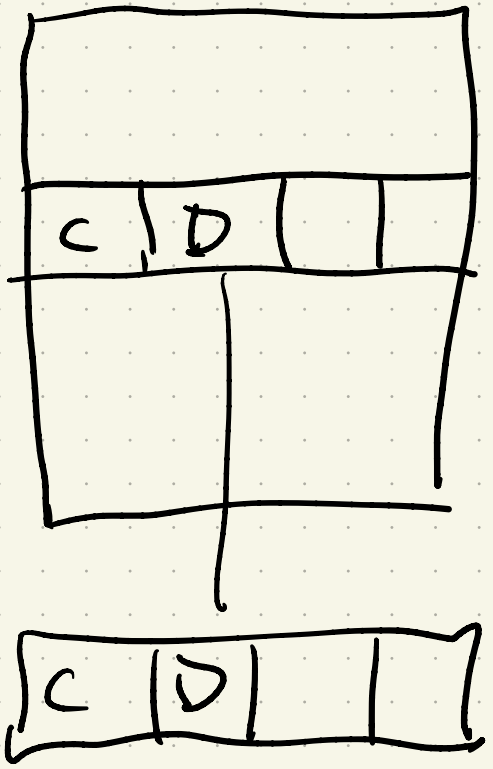
Banking:

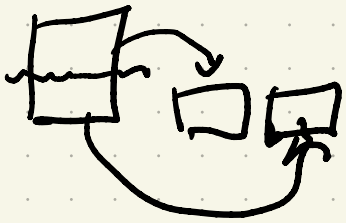
even indexes



Banked sequential

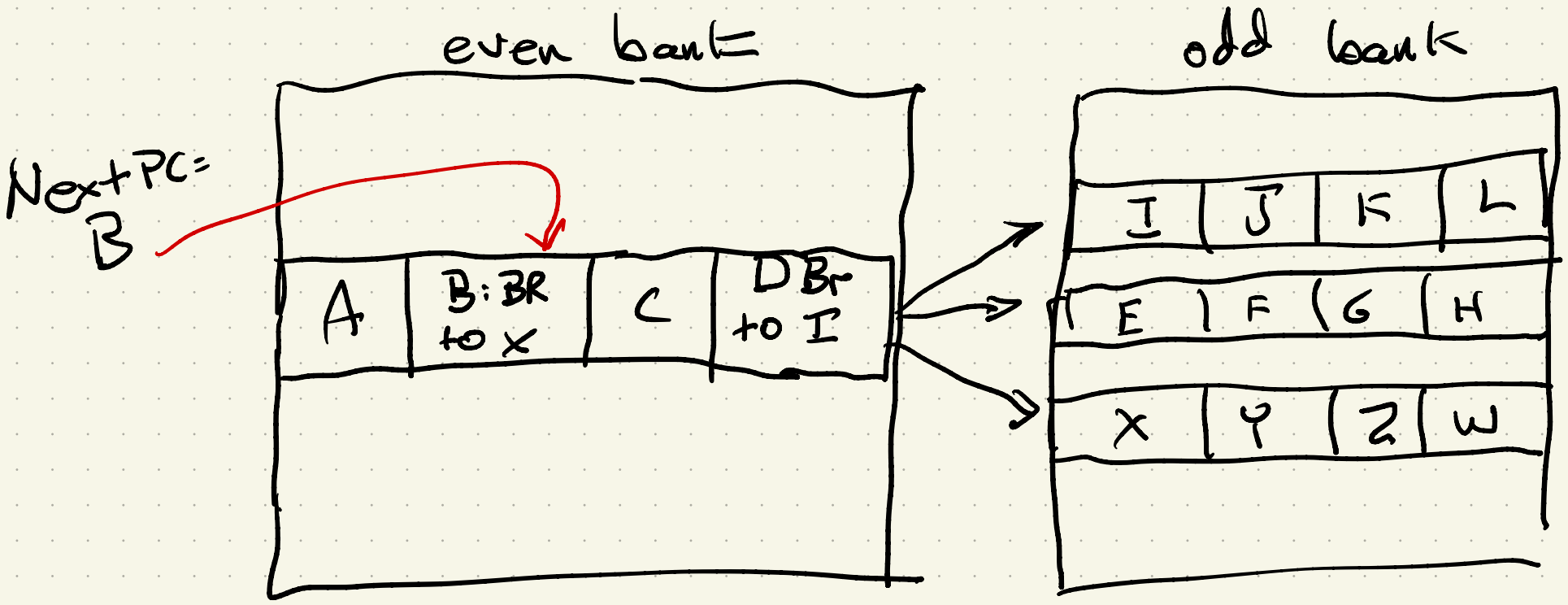
odd indexes





4 inst/cycle

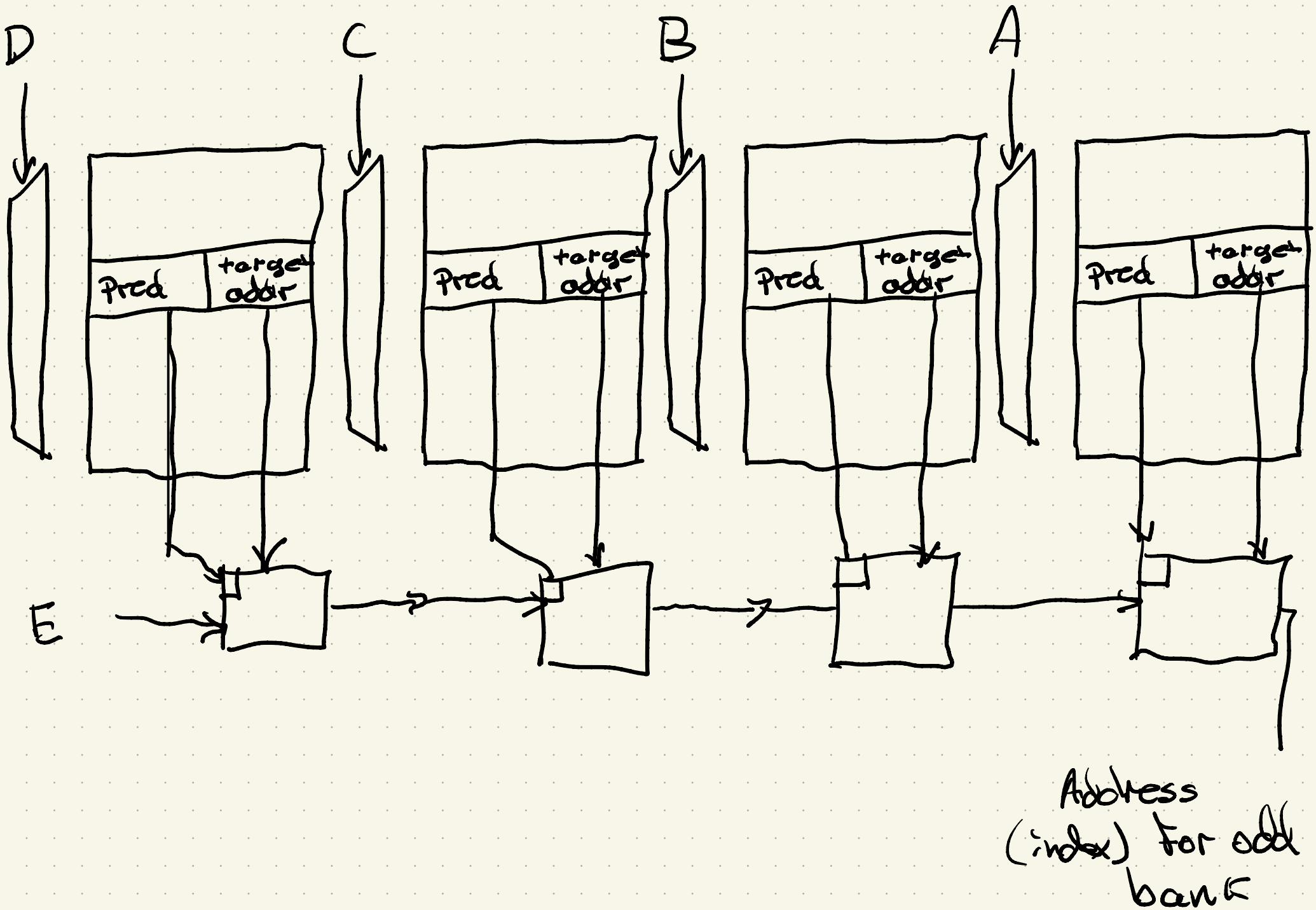
:)

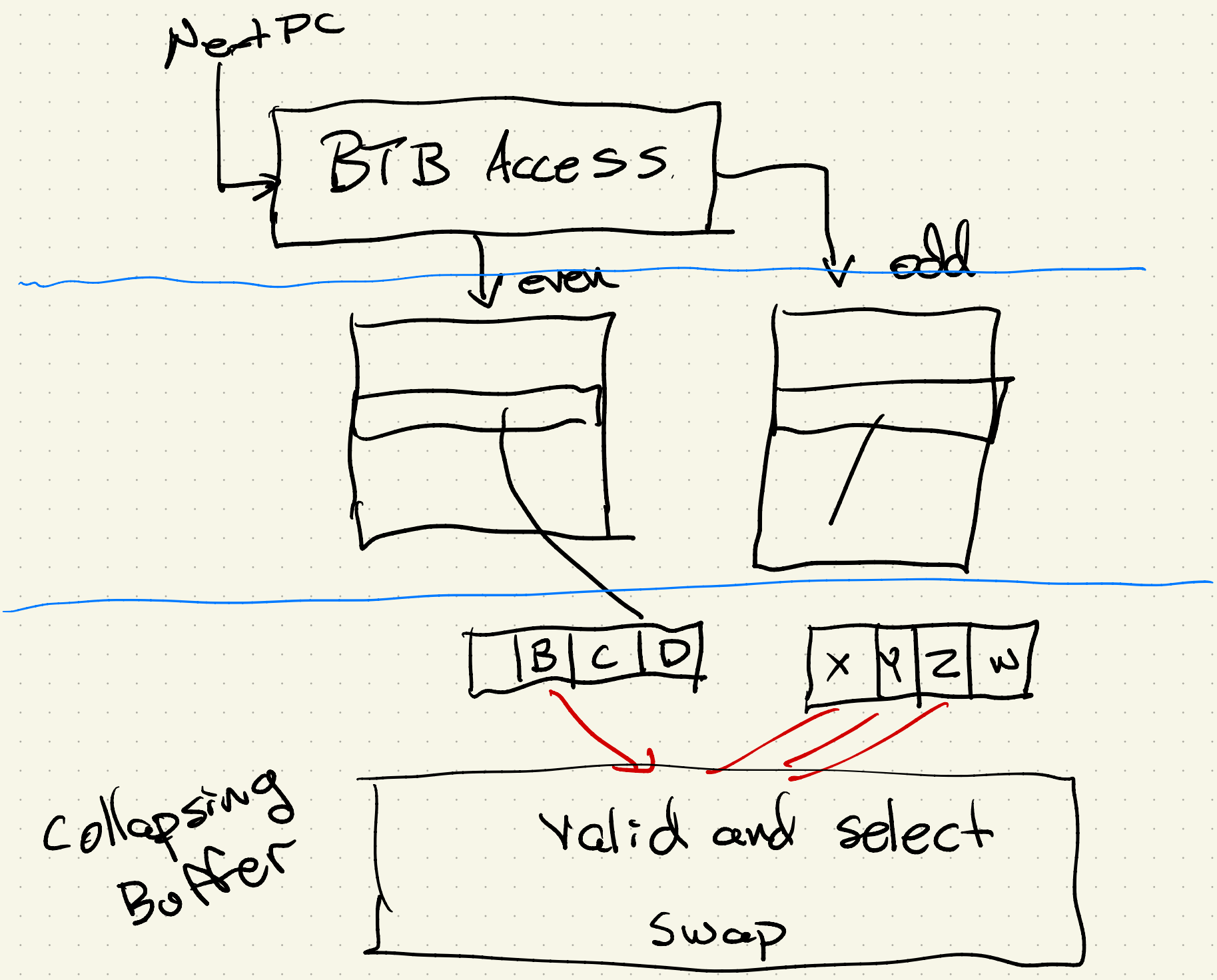


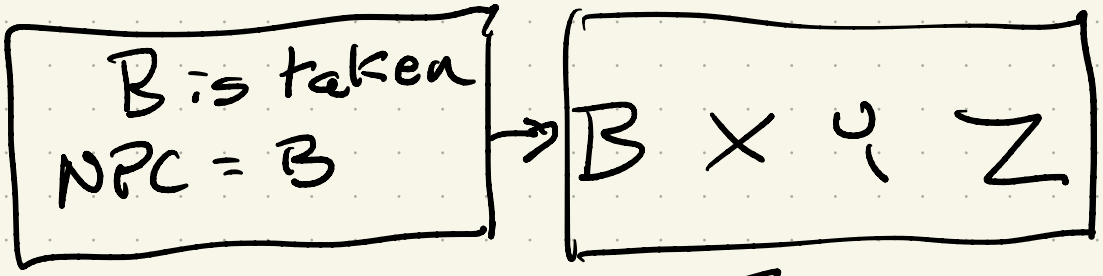
B^{NT} C D^{T} I

B^{T} X Y Z

B^{NT} C D^{NT} E


D
C
B
A
Pred
target addr
Pred
target addr
Pred
target addr
Pred
target addr
E
Address (index) for odd bank

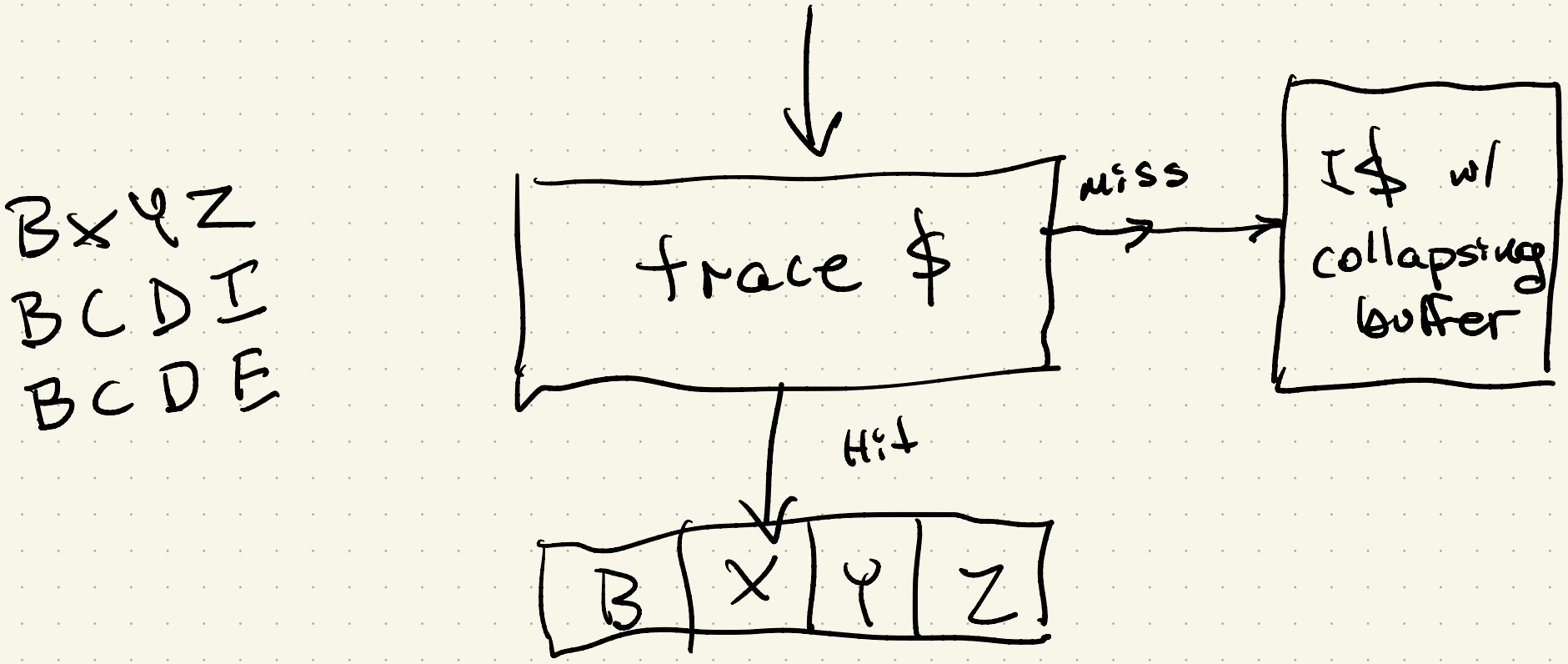

NextPC
BTB Access
even
odd
B
C
D
X
Y
Z
W
Collapsing Buffer
Valid and select
Swap



Trace cache

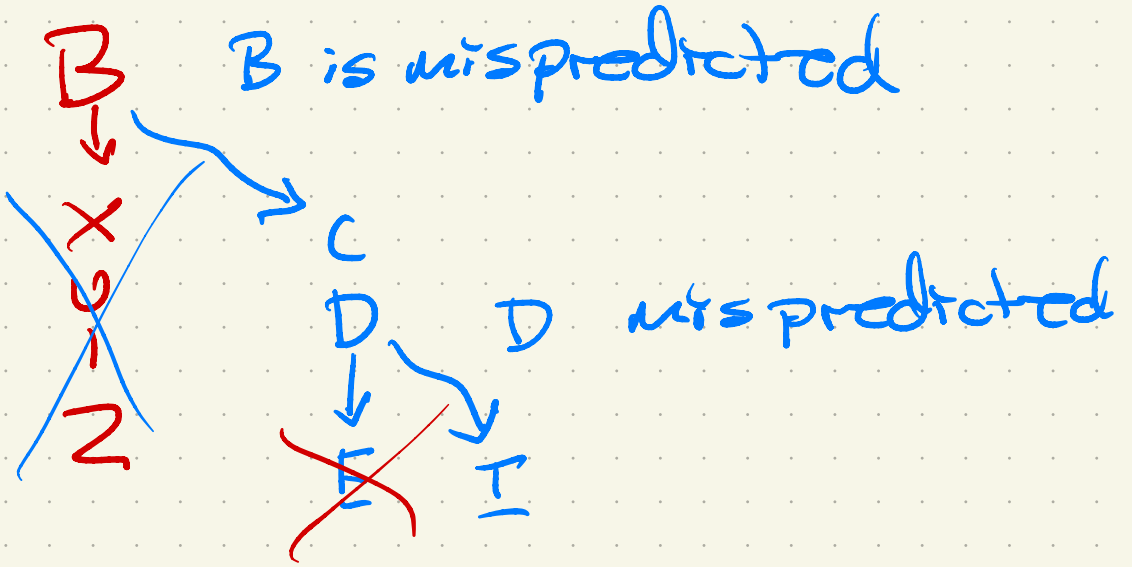
A B C D
0 1 xx

Next PC, Predictions for inst. in the block

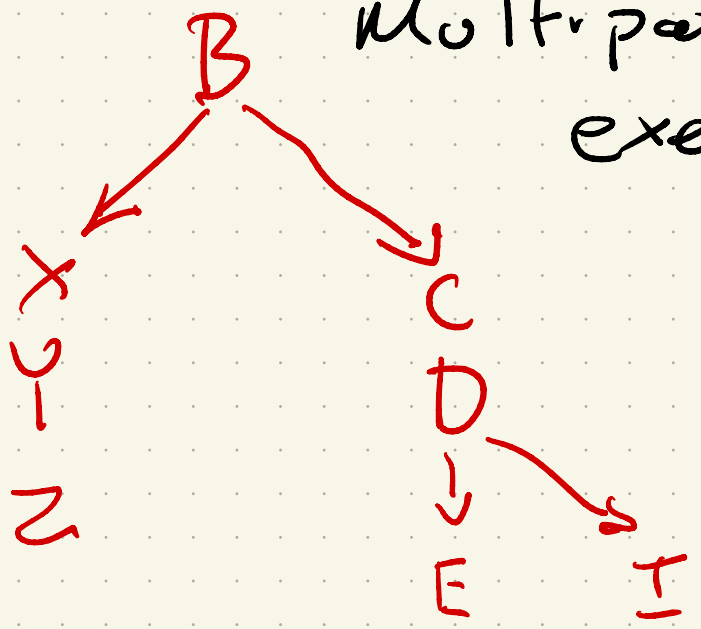


B x Y Z
B C D I
B C D E

Speculative execution



ROB
or
Checkpoint
repair



Multipath execution
execute all paths
undo what is
not needed

Limits of supescalar

Empirically IPC ~ $\log_2$(sched queue size)

No branch prediction $\log_2 5 \sim 2$ IPC

Predict 1 branch ahead

2 IPC + pred.acc · 2 IPC

Complexity of sched queue ~
cycle time and/or power